12 **C**

Using $y = 5\sin(20\pi t)$

At position of maximum displacement, $5 = 5\sin(20\pi t_1)$

$$\Rightarrow 20\pi t_1 = \frac{\pi}{2}$$
$$\Rightarrow t_1 = 0.025\text{s}$$

At position half way between maximum displacement and the equilibrium position,

$$2.5 = 5\sin\left(20\pi t_2\right)$$
$$\Rightarrow 20\pi t_2 = \frac{5\pi}{6}$$
$$\Rightarrow t_2 = 0.042\,\text{s}$$

Required time taken, $t = t_2 - t_1 = 0.042 - 0.025 = 0.017\text{s} = 17\ \text{ms}$

13 **C**